Boosting Hyperspectral Image Classification With Unsupervised Feature Learning

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***Abstract*— The deep learning-based method has shown promising competence in image classification. Its success can be attributed to the ability to learn discriminative feature representation given plenty of labeled data. However, in real-hyperspectral image (HSI) classification applications, since pixel labeling is difficult and costly, the labels we can obtain within an HSI are always limited and noisy (i.e., inaccurate), which consequently causes overfitting of the deep learning-based method. To address this problem, we propose a novel unified deep learning network to employ both labeled and unlabeled data for training, with which the unsupervised structure knowledge, e.g., intracluster similarity and intercluster dissimilarity, inherently contained in those unlabeled data can be exploited to boost the conventional supervised classification. Specifically, we first explore the unsupervised structure knowledge in unlabeled data via a clustering method and formulate a supervised clustering task on those data with the obtained cluster labels. Then, we propose a multitask network to jointly address both the conventional classification task and the formulated supervised clustering task. With a shared feature extraction module and a high-level feature fusion module, the unsupervised structure knowledge contained in unlabeled data can be effectively introduced into the classification task, which is beneficial to learn a more discriminative feature representation and, thus, well mitigates the overfitting problem and improves the classification results. Experimental results on three data sets demonstrate the proposed method can effectively label the unlabeled data within an HSI, especially when the training labels are limited and noisy.**

***Index Terms*— Feature fusion, hyperspectral image (HIS) classification, multitask method, semisupervised learning.**

Manuscript received September 4, 2020; revised January 12, 2021; accepted January 15, 2021. Date of publication March 4, 2021; date of current version December 6, 2021. This work was supported in part by the National Natural Science Foundation of China under Grant 62071387, Grant 61671385, and Grant U19B2037; and in part by the Science, Technology and Innovation Commission of Shenzhen Municipality under Grant JCYJ20190806160210899. *(Corresponding author: Lei Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2021.3054037

I. Introduction

HYPERSPECTAL imaging simultaneously captures spatial and spectral information [1]–[3]. Compared with the traditional images, such as RGB images, which only capture spatial information, the rich spectral information empowers hyperspectral image (HSI) to distinguish different objects of the captured image at the pixel level, which makes HSI widely applied in urban construction and planning, mineral resource exploration, and crop management [4]–[6]. Among these applications, HSI classification is an essential one that aims at assigning a category label to each pixel in the given HSI. Providing only a small amount of pixels in each class, HSI classification can be utilized to assign class labels to the remaining pixels within the HSI. Since the HSI classification method can dramatically decrease the high manual labeling cost, the HSI classification method has long been the research focus [7]–[9], and different HSI classification methods have been proposed. According to the feature utilized, HSI classification methods can be roughly divided into hand-crafted feature-based methods and deep feature-based ones [7].

For hand-crafted feature-based HSI classification methods [10]–[12], feature extraction and classifier design are often considered as two independent ingredients. Among them, the feature extraction method aims at extracting low-dimensional or discriminative features from the original spectrum for better representing each pixel. While classifier focuses on developing strategies to separate pixels of one class from the others [13]–[16]. To date, amounts of hand-crafted feature-based HSI classification methods have been proposed from different perspectives. For example, Demir and Erturk [17] utilized the original spectrum to represent each pixel directly and then trained a support vector machine (SVM) classifier on them. Jia *et al.* [18] extracted the Gabor features for hyperspectral classification. Qian *et al.* [16] proposed to extract 3-D discrete texture features by wavelet transformation and implemented pixelwise classification based on a structured sparse logistic regression scheme. Li *et al.* [19] proposed a structure-aware collaborative representation method with the Tikhonov regularization for HSI classification. Zhang *et al.* [20] further used multiple features to improve the classification results. To eliminate the

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redundant information within the spectrum, Yuan *et al.* [21] proposed to select the optimal subset of bands using the graph model to mine the structure between bands. However, the feature extracted by the above-mentioned methods is always based on some predefined criteria independent of the classifier, which makes it difficult to seamlessly fit the classier. More seriously, the shallow structure of the feature extraction methods impedes them to extract discriminative enough feature for accurate classification.

In contrast with the hand-crafted feature-based HSI classification method, the deep feature-based method has the merit of extracting deep features. It also integrates feature extraction and classifier design into a unified framework, with which discriminative feature can be obtained, and thus, results in more accurate HSI classification results [22]–[25]. Different deep learning-based methods are proposed accordingly. For example, Chen *et al.* [22] introduced a stacked auto-encoder to extract the deep feature representation for each pixel, and then predicted their labels via backpropagation under supervision. Ma *et al.* [26] proposed to utilize a deep auto-encoder network to extract spectral–spatial features and then applied a collaborative representation-based classifier for HSI classification. Witnessing the success of convolutional neural network (CNN) in various computer vision tasks, more and more deep HSI classification methods are constructed with the CNN structure. For example, Chen *et al.* [8] employed a 3-D convolutional layer to extract deep spectral–spatial features and strategies, such as ℓ_2 regularization and dropout, were utilized to avoid overfitting. Zhou and Wei [24] proposed to combine the spectral feature learning and spatial feature learning in a hierarchical fashion to exploit both discriminative spectral and spatial information. Since it is difficult to train a deep neural network when the network goes deeper due to the gradient vanishing problem, He *et al.* [27] proposed a residual learning structure to address this problem. Following this idea, recent HSI classification networks also exploit residual network for HSI classification. For example, Paoletti *et al.* [28] proposed an end-to-end residual-based pyramid network-based HSI classification method, in which the raw 3-D cubes were adopted as input and residual blocks are utilized to connect different 3-D convolutional layer through the identity mapping for HSI classification.

Despite the advantages of existing deep feature-based methods, their success relies on plenty of labeled training data that are provided beforehand. However, within an HSI, it is hard to ask experts to label a large number of pixels since labeling HSI data are difficult and costly. In such a case, these deep feature-based methods are prone to overfitting and failing to learn discriminative enough feature representation [29]. Thus, it is necessary to investigate how to adapt the HSI classification method to the case when labels are few and noisy [30].

Some methods have been proposed to address this problem. One category of methods attempt to generate more labeled data based on the labeled training data, including data augmentation strategy-based method [31] and pixel-pair strategy-based one [32]. Though these methods can generate more labeled data, they rely severely on the existing labeled data. Once the labeled data is severely limited (i.e., only a few labeled samples) or the given labels are noisy, the generated labeled data are still insufficient to overcome the overfitting problem, or easily mislead the classifier. Another category of methods focuses on finding a suitable way to use the unlabeled data to regularize the supervised learning on labeled training data since there are plenty of unlabeled data in real applications [33]. They are also termed semisupervised methods. For example, Yang *et al.* [34] proposed a spatiospectral Laplacian SVM (SS-LapSVM) method for HSI classification. SS-LapSVM simultaneously utilizes a clustering assumption on spectral vectors and neighborhood spatial constraints to separately formulate a manifold regularizer and a spatial regularizer, with which the original supervised SVM method is regularized and the classification accuracy is improved. Though such a category of methods has been explored for hand-crafted feature-based methods, the majority of these methods cannot be directly applied to deep feature-based ones since the hand-crafted feature-based method demands a small number of labeled training samples to extract discriminative features, whereas the deep feature-based method requires more. Thus, how to exploit the unlabeled data for the deep learning-based HSI classification method is still a challenging problem.

To address this problem, we propose a new unified supervised learning network to jointly take the advantages of both the labeled and the unlabeled data to label data within the HSI. To guarantee training the proposed network with the unlabeled data in a supervised way, we apply a clustering method on the given HSI data beforehand and take the resultant cluster labels as the supervision for those unlabeled pixels to formulate a supervised clustering task. We then propose a multitask supervised network, which simultaneously accomplishes the formulated supervised clustering task and the conventional classification task. In the proposed network, these two tasks share the same basic feature extraction module. A feature fusion module is also developed to further fuse the high-level feature from two tasks for better feature representation. By doing these, the unsupervised structure knowledge, such as intracluster similarity and intercluster dissimilarity contained in both unlabeled and labeled data can be automatically exploited to regularize the classification task, which enables the learned feature to be discriminative and robust to noise, and thus, obviously improves the classification accuracy within the HSI. Experimental results on three data sets demonstrate the impressive performance of the proposed method, especially when the label is limited and noisy.

In summary, the main contribution of this study can be summarized as follows.

1) We propose to utilize the unsupervised structure knowledge inherently contained in the unlabeled data to regularize the conventional supervised classification and improve the classification result within the HSI, which is beneficial to learn discriminative feature representations given only a few labeled data or noisy labels.
2) We propose a multitask supervised network, which naturally incorporates the labeled and unlabeled data into a unified framework. In addition, the learned unsupervised knowledge can well regularize the classification task

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benefiting from the specifically designed shared feature extraction module and the feature fusion module.

3) We report significant advantages over other HSIs classification methods on three benchmark data sets for classification within an HSI, especially given only a few labeled data or noisy labels.

II. Proposed Method

For HSI classification, one of the important applications is to label the pixels within an HSI providing only a few labeled pixels, since HSI classification can dramatically reduce the labeling cost. Specifically, providing an HSI, experts only need to label a small number of samples within the HSI, and then utilize the HSI classification method based on these labeled data to automatically label the remaining pixels within the same HSI.

Denote the labeled data set by experts as $\mathcal{D}_l = \{\boldsymbol{x}_i, \boldsymbol{y}_i\}_{i=1}^{M}$ and the unlabeled data set within the HSI as $\mathcal{D}_u = \{\boldsymbol{x}_i\}_{i=M+1}^{M+N}$, a commonly utilized strategy is to learn the relationship between the pixel and its class label via the labeled data set $\mathcal{D}_l$, and then utilize the learned relationship to predict the label for the unlabeled data set $\mathcal{D}_u$, i.e., assign each pixel $\boldsymbol{x}_t$ a class label. $\boldsymbol{x}_i \in \mathbb{R}^{d\times 1}$ represents a d-dimensional sample (i.e., pixel) and $\boldsymbol{y}_i \in [1, 2, \ldots, L]$ is the label of $\boldsymbol{x}_i$. M, L, and N represent the number of labeled samples, classes (i.e., categories), and unlabeled samples, respectively. Taking advantages of deep neural network in extracting deep features and integrating feature extraction and classifier into an end-to-end framework, the deep learning-based method dominates the way formulating the relationship between $\boldsymbol{x}_i$ and $\boldsymbol{y}_i$. However, the conventional deep neural network-based method always demands plenty of accurately labeled training samples, whereas the obtained labels are always few (i.e., M is small) and inaccurate (i.e., some pixels are mistakenly labeled into other categories) since pixel labeling is difficult, costly, and only can be accomplished by experts. Once applied these labeled samples for classification, the method is prone to be overfitting. Thus, it is essential to investigate how to adapt the deep neural network-based method to samples with a few and inaccurate labels. Although only a few labeled samples can be provided, plenty of unlabeled samples are available within the HSI to be classified. These unlabeled samples can provide effective constraints to regularize the overfitting problem, which promotes us to propose a novel deep learning network to employ both labeled and unlabeled data for labeling the pixels within the HSI. In the following, we first present the proposed network structure and then detailed introduce each module with the proposed network.

A. Framework of the Proposed Method

In this study, we propose to utilize the unlabeled data to regularize the overfitting problem when a small number of labeled samples are provided for training a network structure. Thus, how to utilize the unlabeled data becomes the focus of this study. Considering unsupervised structure knowledge, such as data similarity inherently contained in the unlabeled samples, we explore the unsupervised structure in $\mathcal{D} = \{\mathcal{D}_l, \mathcal{D}_u\}$ (without using the label $\boldsymbol{y}_i$ of $\boldsymbol{x}_i$) via a clustering method and then formulate the supervised clustering task on those data with the obtained cluster labels. It is noticeable that we use $\boldsymbol{c}_i$ to represent cluster label in order to discriminate it from the class label $\boldsymbol{y}_i$ in this study. The merit of utilizing the clustering method to generate the cluster label has two merits. First, the resulted cluster labels can reflect the intracluster similarities and the intercluster dissimilarities, which can be utilized to represent the unsupervised structure. Second, with the resulted cluster labels, the original clustering problem can be cast into a supervised task, with which a large number of unlabeled samples can be easier incorporated for a supervised task.

Based on the generated clustering label, we construct a multitask classification network, as shown in Fig. 1. It can be seen that the structure of the proposed multitask network consists of a conventional classification task and a supervised clustering task. Within the proposed network, the proposed shared feature extractor module and the feature fusion module play an important role to regularize the overfitting problem with the unlabeled data.

After extracting elementary features on all samples $\{\boldsymbol{x}_i\}_{i=1}^{M+N}$ via the shared feature extraction module F_θ, the obtained features are then used for these two tasks. It is noticeable that the labeled data from $\{\boldsymbol{x}_i\}_{i=1}^{M}$ has the class label $\boldsymbol{y}_i$ and the cluster label label $\boldsymbol{c}_i$ since the labeled data are also utilized for clustering. Thus, the obtained features from $\{\boldsymbol{x}_i\}_{i=1}^{M}$ via F_θ are fed into the conventional classification task and the supervised clustering task. To guarantee the obtained features suitable for both the conventional classification task and the supervised clustering task, the features from the shared feature extraction module have to adjust themselves, which makes the introduced supervised clustering task highly correlated with the conventional classification task. As a result, the unlabeled data set $\mathcal{D}_u = \{\boldsymbol{x}_i\}_{i=M+1}^{M+N}$ can better regularize the overfitting problem when extracting features via the shared feature extraction module.

In addition, considering the feature extracted from the high-level feature extractors F_u (for the supervised clustering task) and F_s (for the conventional classification task) shares the same class label for the labeled data $\boldsymbol{x}_i$, we further propose a feature fusion model to fuse two kinds of high-level features obtained from the same labeled data. To guarantee the fused feature can well predict its class label, the high-level feature obtained from F_u have to be adjusted, which will also make the unlabeled data set $\mathcal{D}_u$ better regularize the overfitting problem, and thus, obtain better HSI classification results. In the following, we will introduce each module within the proposed network in detail.

B. Shared Feature Extractor

The proposed network consists of a conventional classification task and a supervised clustering task. For these two tasks, the cluster labels essentially reflect the data similarities within the HSI data while the predefined class labels represent the physical attributes of samples. Due to the difference between the class labels and the cluster labels, suitable representation is required to bridge these two tasks. We, thus, model the

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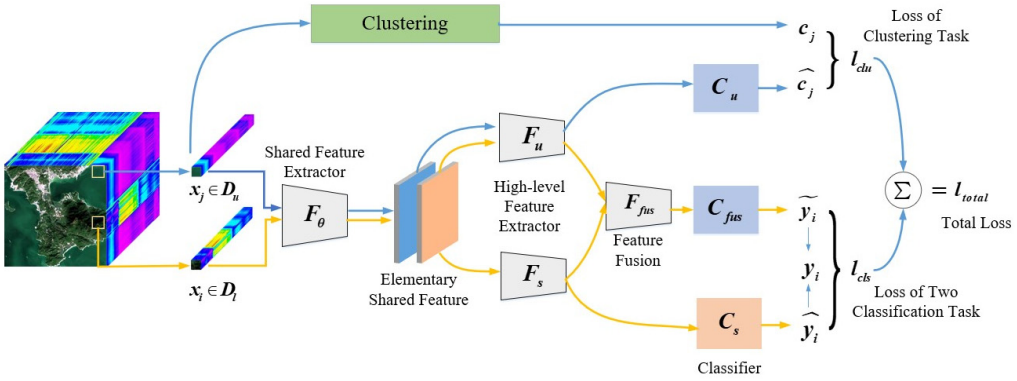


Fig. 1. Structure of the proposed network.

shared feature extraction module for this purpose, which has the following two advantages. First, with the shared feature extraction module, large amounts of unlabeled data can be introduced, with which the overfitting problem (when using the labeled data only) can be well regularized. Second, the shared feature extraction module aims at learning features that can adapt to both the classification and clustering tasks, which will result in the features obtained from the clustering task discriminative, i.e., the cluster label can reflect the class property to some extent especially with a large cluster number (see the ablation study in experimental part for details).

In this study, CNN is utilized to extract the elementary shared feature from the HSI data. To capture both local spatial and spectral information, 3-D CNN is adopted. Specifically, we construct a 3-D CNN F_θ consisting of ten convolutional layers. Each layer is composed of a 3-D convolutional component, a batch normalization component together with a rectified linear unit (ReLU) activation function [35]. The convolutional component with a stride focuses on extracting the local information within the spectrum. The settings, including the feature number, kernel size, and stride of each convolutional layer, are summarized in Table I. Taking the first convolutional layer as an example, 64 different 3-D convolution kernels with size $3 \times 1 \times 3$ are used to extract different kinds of spectral and local spatial feature patterns. Each kernel is used accomplished with stride size $1 \times 1 \times 3$ to reduce the spectral dimension of the HSI in the feature space. A similar structure as the first convolutional layer is adopted from layer 2 to layer 10, in which the output from $(n-1)$th layer is used as the input of nth layer. The only difference between the first layer and those layers is the numbers used for the convolution kernel, the feature map, and the stride. It is noticeable that with the adopted settings, the final layer outputs a feature vector.

TABLE I
STRUCTURE OF THE SHARED FEATURE EXTRACTOR

Layer	Input size	Feature number	Kernel size	stride
1	p	64	$3 \times 1 \times 3$	(1, 1, 3)
2	D_1	64	$1 \times 1 \times 3$	(1, 1, 1)
3	D_1	64	$1 \times 3 \times 3$	(1, 1, 1)
4	D_1	128	$1 \times 1 \times 3$	(1, 1, 2)
5	D_2	128	$3 \times 1 \times 3$	(1, 1, 1)
6	D_2	128	$1 \times 1 \times 3$	(1, 1, 1)
7	D_2	256	$1 \times 3 \times 3$	(1, 1, 2)
8	D_3	256	$1 \times 1 \times 3$	(1, 1, 1)
9	D_3	256	$1 \times 1 \times 2$	(1, 1, 1)
10	D_3	512	$1 \times 1 \times D_3$	(1, 1, 1)

C. High-Level Global Feature Extractor

Considering the elementary features are obtained by stacking local results from the local convolutional kernels, they are insufficient to represent the global information. Thus, we further construct high-level global feature extractor modules, including F_s and F_u, to capture global and deeper information from the obtained elementary features, which can better reflect the characteristics for classification or clustering. Specifically, we construct the high-level feature extraction module via a fully connected layer, which includes a fully connected component, a batch normalization component, and a ReLU activation function. Since the fully connected component introduces the mutual-correlations among each element of the elementary features, the global features regarding the spectra are generated.

The difference between F_s and F_u lies in the input data. Only the elementary features from labeled data are used for F_s, while the features from both labeled data $\mathcal{D}_l$ and unlabeled data $\mathcal{D}_u$ are utilized for F_u. It is noticeable that the obtained global features from F_u are then fed into feature fusion module F_{fus} and classifier module C_u, respectively.

D. Feature Fusion

Considering the feature extracted from the high-level feature extractors F_u (for the supervised clustering task) and F_s (for the conventional classification task) shares the same class label for the labeled data $\boldsymbol{x}_i$, feature fusion module F_{fus} is built to enhance the discriminability of the obtained feature from the unlabeled data. Specifically, in the feature fusion module, two kinds of high-level global features obtained by the labeled data $\mathcal{D}_l$ are fused, namely, the feature output by the high-level global feature extractor F_s and that from F_u are fused. Since those two kinds of features are extracted from the same labeled

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data, we can use the true class label to train the network. To obtain a good classification result, the features from F_u have to adjust themselves during the training stage to predict the class label (i.e., suitable for the classification task), which, thus, can further influence the shared feature extraction module to generate discriminative features from the data $\mathcal{D}_u$ with the backpropagation algorithm.

Denote the features obtained from F_s and F_u as vectors $\boldsymbol{f}_s$ and $\boldsymbol{f}_u$, respectively, we can fuse them as

$$\boldsymbol{f}_{\text{fus}} = F_{\text{fus}}(\boldsymbol{f}_u, \boldsymbol{f}_s). \tag{1}$$

F_{fus} denotes a general representation of fusion strategy. In this study, six strategies, including $F_{\text{fus}}^{\text{conc}}$, $F_{\text{fus}}^{\text{att}}$, $F_{\text{fus}}^{\text{tra}}$, $F_{\text{fus}}^{\text{mul}}$, $F_{\text{fus}}^{\text{conv}}$, and $F_{\text{fus}}^{\text{bil}}$ are explored for fusing the features, which are listed as follows.

1) Concatenation $F_{\text{fus}}^{\text{conc}}$

$$\boldsymbol{f}_{\text{fus}} = F_{\text{fus}}^{\text{conc}}(\boldsymbol{f}_u, \boldsymbol{f}_s) = [(\boldsymbol{f}_u)^T, (\boldsymbol{f}_s)^T]^T. \tag{2}$$

T represents transposition operation.

2) Attention $F_{\text{fus}}^{\text{att}}$

$$\boldsymbol{f}_{\text{fus}} = F_{\text{fus}}^{\text{att}}(\boldsymbol{f}_u, \boldsymbol{f}_s) = \boldsymbol{f}_u + \boldsymbol{f}_s \otimes \text{Att}(\boldsymbol{f}_u) \tag{3}$$

$\otimes$ represents dot production. Att$(\cdot)$ denotes the operation accomplished via a fully connected layer together with a sigmoid activation function, which assigns different weights to $\boldsymbol{f}_s$ according to $\boldsymbol{f}_u$.

3) Transformation $F_{\text{fus}}^{\text{tra}}$

$$\boldsymbol{f}_{\text{fus}} = F_{\text{fus}}^{\text{tra}}(\boldsymbol{f}_u, \boldsymbol{f}_s) = \boldsymbol{f}_s \otimes \text{Scale}(\boldsymbol{f}_u) + \text{Bias}(\boldsymbol{f}_u). \tag{4}$$

Scale$(\cdot)$ denotes the operation accomplished via a fully connected layer together with a sigmoid activation function. Bias$(\cdot)$ represents the operation using a fully connected layer.

4) Multiplication $F_{\text{fus}}^{\text{mul}}$

$$\boldsymbol{f}_{\text{fus}} = F_{\text{fus}}^{\text{mul}}(\boldsymbol{f}_u, \boldsymbol{f}_s) = \boldsymbol{f}_s \otimes \boldsymbol{f}_u. \tag{5}$$

5) Convolution $F_{\text{fus}}^{\text{conv}}$

$$\boldsymbol{f}_{\text{fus}} = F_{\text{fus}}^{\text{conv}}(\boldsymbol{f}_u, \boldsymbol{f}_s) = \text{Conv}([(\boldsymbol{f}_u)^T, (\boldsymbol{f}_s)^T]^T). \tag{6}$$

Conv$(\cdot)$ denotes a 1-D convolutional operation.

6) Bilinear $F_{\text{fus}}^{\text{bil}}$

$$\boldsymbol{f}_{\text{fus}} = F_{\text{fus}}^{\text{bil}}(\boldsymbol{f}_u, \boldsymbol{f}_s) = \text{Flatten}((\boldsymbol{f}_u)^T \times \boldsymbol{f}_s). \tag{7}$$

Flatten$(\cdot)$ represents flatten operation, which outputs a feature vector from a matrix.

E. Classifier

Classifiers, including C_s, C_u, and C_{fus} are used to classify each high-level global feature into different class or cluster. To accomplish this goal, a fully connected layer that contains a fully connected component and a softmax function is exploited to output a nonnegative vector. The dimension of the vector equals the class number and adopts a probabilistic representation (i.e., the sum of all elements in the vector equals 1). Specifically, for classifiers C_s and C_{fus}, the dimension of the output vector equals L. While for the classifier C_u, the dimension of the output vector equals the cluster number K.

To obtain the cluster labels in advance, we utilize the K-means clustering method on both $\mathcal{D}_u$ and $\mathcal{D}_l$. The number of cluster K for the clustering method will influence the performance of the proposed method. The relative experiments and analysis can be seen from the ablation study in Section IV.

F. Training

The above-mentioned introduced modules constitute the entire end-to-end network. To optimize such a network, we first need to define a suitable loss function and then minimize the loss function. Considering the entire network contains two tasks, including a conventional classification task on $\mathcal{D}_l$ and a supervised clustering task on $\mathcal{D}_u$, we first separately define the loss function for each task, and then obtain the overall function for the entire network.

For the supervised clustering task on $\mathcal{D}_u$, we adopt the cross entropy loss defined as

$$l_{\text{clu}} = -\sum_{i=M+1}^{M+N} \boldsymbol{c}_i \log(\hat{\boldsymbol{c}}_i) \tag{8}$$

where $\hat{\boldsymbol{c}}_i$ is the predicted cluster label of $\boldsymbol{x}_i$ from the classifier module C_u as

$$\hat{\boldsymbol{c}}_i = C_u(\boldsymbol{f}_u^i); \quad \boldsymbol{f}_u^i = F_u(\boldsymbol{f}_\theta^i); \quad \boldsymbol{f}_\theta^i = F_\theta(\boldsymbol{x}_i). \tag{9}$$

$\boldsymbol{f}_\theta^i$ and $\boldsymbol{f}_u^i$ represent the obtained feature from the shared feature extractor module F_θ and the high-level global feature extractor module F_u, respectively.

For the conventional classification task on $\mathcal{D}_l$, both the supervised feature and the fusion feature are utilized to predict the class label, with which the loss function is defined as

$$l_{\text{cls}} = -\sum_{i=1}^{M} (\boldsymbol{y}_i \log(\hat{\boldsymbol{y}}_i) + \lambda \boldsymbol{y}_i \log(\widetilde{\boldsymbol{y}}_i)). \tag{10}$$

λ is a balance coefficient. $\hat{\boldsymbol{y}}_i$ and $\widetilde{\boldsymbol{y}}_i$ represent the predicted labels from the supervised feature and fusion feature, respectively, which can be calculated by

$$\hat{\boldsymbol{y}}_i = C_s(\boldsymbol{f}_s^i); \quad \boldsymbol{f}_s^i = F_s(\boldsymbol{f}_\theta^i); \quad \boldsymbol{f}_\theta^i = F_\theta(\boldsymbol{x}_i) \tag{11}$$

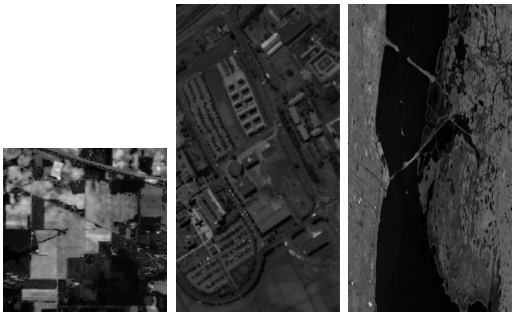
$$\widetilde{\boldsymbol{y}}_i = C_{\text{fus}}(\boldsymbol{f}_{\text{fus}}^i); \quad \boldsymbol{f}_{\text{fus}}^i = F_{\text{fus}}(\boldsymbol{f}_u^i, \boldsymbol{f}_s^i)$$
$$\boldsymbol{f}_u^i = F_u(\boldsymbol{f}_\theta^i); \quad \boldsymbol{f}_s^i = F_s(\boldsymbol{f}_\theta^i); \quad \boldsymbol{f}_\theta^i = F_\theta(\boldsymbol{x}_i). \tag{12}$$

To simultaneously satisfy above-mentioned two tasks, the following joint loss is finally defined based on above-mentioned two loss functions as:

$$\begin{aligned} l_{\text{total}} &= \lambda_1 l_{\text{cls}} + \lambda_3 l_{\text{clu}} \\ &= -\Bigg(\lambda_1 \sum_{i=1}^{M} \boldsymbol{y}_i \log(\hat{\boldsymbol{y}}_i) + \lambda_2 \sum_{i=1}^{M} \boldsymbol{y}_i \log(\widetilde{\boldsymbol{y}}_i) \\ &\qquad + \lambda_3 \sum_{i=M+1}^{M+N} \boldsymbol{c}_i \log(\hat{\boldsymbol{c}}_i)\Bigg). \end{aligned} \tag{13}$$

λ_1, λ_2, and λ_3 are predefined scale factors to balance the contributions from three loss functions, in which $\lambda_2 = \lambda \times \lambda_1$. It is noticeable that all labels, including the class labels and the

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(a) (b) (c)

Fig. 2. (a) Picture of the 24th band of Indian Pines data set, (b) 69th band of PaviaU data set, and (c) 71th band of KSC data set.

cluster labels, are based on one-hot representation for training. With such a loss function, we then train the proposed network via the backpropagation algorithm.

After training, the test process is rather simple. When we feed the unlabeled data within the HSI into the network, the output result obtained from the conventional classification task is the predicted class label.

III. Experiments and Analysis

In this section, we conduct plenty of experiments in terms of classification accuracy and the robustness to label noise, to demonstrate the effectiveness of the proposed method comparing with other competing methods.

A. Hyperspertral Data Sets

In this study, three HSI benchmark data sets, including the Indian Pines data set, Pavia University (PaviaU) data set, and Kennedy Space Center (KSC) data set are adopted to verify the performance of proposed method, as shown in Fig. 2.

1) *Indian Pines Data Set:* The Indian Pines Data set is collected by an airborne visible/infrared imaging spectrometer (AVIRIS) sensor over the Indian Pines test site in northwestern Indiana, USA. The captured HSI contains 145×145 pixels with 200 meters spatial resolution and 220 spectral reflectance bands ranging from 0.4 to 2.5 μm. There are 16 classes in total in this data set. In experiments, we remove 20 water-absorption bands and keep the remaining 200 bands for experiments. To make the labeled data in each class roughly balanced, we choose nine classes from 16 classes for experiments, which have 9234 labeled samples in total.
2) *PaviaU Data Set:* The PaviaU data set is collected via the reflective optics system imaging spectrometer sensor over the University of Pavia, Pavia, Italy, with 1.3-m spatial resolution. The captured PaviaU data set has 610×340 pixels, nine classes, and 103 spectral reflectance bands ranging from 0.43 to 0.86 μm. In experiments, all classes and bands are utilized for classification.
3) *KSC Data Set:* The KSC Data set is acquired by the AVIRIS sensor over the Kennedy Space Center, Merritt Island, FL, USA. The data have 512×614 pixels, 13 classes, and 224 bands ranging from 0.4 to 2.5 μm. In experiments, we remove water-absorption bands and low signal-noise-ratio bands and keep the remaining 176 bands for experiments.

TABLE II
Batchsize Settings With Different Number of Labeled Data

Number of Labeled Samples	5	10	15	20	50
Batch size of Classification Task	20	40	64	80	100
Batch size of Supervised Cluster Task	40	80	100	100	100

B. Comparison Methods and Measures

To demonstrate the superioity of the proposed method, we compare it with the competing methods, including SVM, 3D-CNN, SS-LapSVM [34], pyramid residual network (PRN) [28], SpecPatConvnet [36], and deep cube-pair network (DCPN) [37]. SVM is a conventional hand-crafted feature-based method, which specializes in dealing with a small sample problem. SpecPatConvnet and PRN are two state-of-the-art deep learning-based methods utilizing both spatial and spectral information for classification. SpecPatConvnet adopts a CNN structure [36], while PRN utilizes a residual network structure. SS-LapSVM is a semisupervised method utilizing both the labeled samples and the unlabeled samples. DCPN is a cube-pair-based deep learning-based method, which adopts a pair strategy to generate more labeled samples from the existing labeled data. We term the proposed method as Ours in this study. Considering the proposed method includes both classification tasks and clustering tasks, we model a variant method of Ours, which adopts the same structure as Ours but only preserves the classification branch. We term this variant method as 3D-CNN, which is a supervised method utilizing the labeled data only.

To evaluate the performance of classification accuracy, we utilize three conventional evaluation metrics, including overall accuracy (OA), the Kappa coefficient, and average accuracy (AA). Larger OA, Kappa, and AA indicate better classification performance.

C. Experimental Settings

In this study, the proposed method is realized in the TensorFlow platform on a computer with Intel Xeon CPU E5-2650 v4 4.20 GHz, 64-GB DDR4 RAM, and a GPU NVIDIA GeForce GTX 1080Ti. During training, the learning rate is set as 1×10^{-3} initially and exponentially decayed with decay factor 0.9 every 10 000 iterations. The maximum iteration number is set as 200 000. In experiments, the batch size is set related to the number of labeled samples, which is summarized in Table II. For a fair comparison, similar to PRN and SpecPatConvnet, we utilize a local cube centered at a chosen pixel instead of that pixel for training in order to utilize both spatial and spectral information. In experiments, the spatial size of the local cube is set as 5×5, and all spectral dimensionality is utilized for the proposed method. The comparison methods, including PRN and SpecPatConvnet,

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TABLE III

CLASSIFICATION RESULTS OF ALL METHODS PROVIDING TEN LABELED TRAINING SAMPLES PER CLASS ON INDIAN PINES DATA SET (%). BEST RESULTS ARE IN BOLD. THE NUMBERS IN PARENTHESES ARE STANDARD VARIANCES (%)

Class	SVM	3D-CNN	PRN	SpecPat Convnet	SS-LapSVM	DCPN	Ours
1	41.94	23.22	46.29	60.81	39.76	40.63	**93.61**
2	70.51	47.17	57.69	70.51	66.92	68.58	**99.74**
3	76.90	90.30	58.42	69.97	86.83	73.90	**99.53**
4	51.76	90.88	49.96	64.70	44.28	47.21	**99.70**
5	81.77	88.25	90.65	95.32	77.10	78.03	**99.53**
6	64.20	38.14	73.53	76.78	55.31	61.82	**99.34**
7	48.72	30.15	62.68	67.02	42.98	46.69	**90.88**
8	75.60	80.67	58.72	77.48	71.29	74.29	**99.43**
9	71.02	65.02	76.95	83.53	76.21	69.13	**99.58**
OA	$59.30_{(1.16)}$	$50.63_{(3.26)}$	$62.71_{(1.39)}$	$71.70_{(2.39)}$	$57.84_{(1.73)}$	$58.29_{(2.15)}$	$\mathbf{96.30}_{(1.25)}$
AA	$64.71_{(1.74)}$	$61.53_{(2.76)}$	$63.88_{(1.28)}$	$74.01_{(2.45)}$	$62.30_{(2.64)}$	$62.25_{(3.04)}$	$\mathbf{97.93}_{(1.16)}$
$\kappa \times 100$	$53.85_{(1.25)}$	$44.78_{(2.28)}$	$57.54_{(1.29)}$	$67.87_{(1.83)}$	$52.67_{(2.25)}$	$53.13_{(2.15)}$	$\mathbf{95.68}_{(1.86)}$

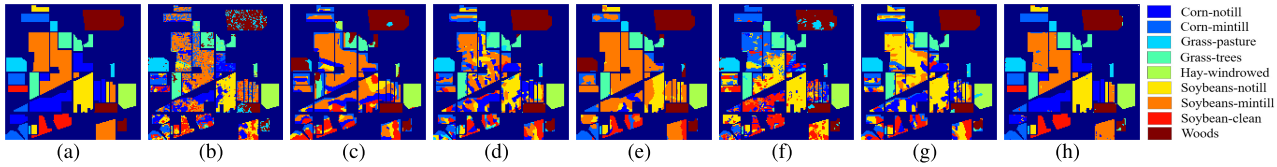


Fig. 3. Classification maps of different methods providing ten labeled training samples per class on Indian pines data set. (a) Ground truth. (b) SVM. (c) 3D-CNN. (d) PRN. (e) SpecPatConvnet. (f) SS-LapSVM. (g) DCPN. (h) Ours.

are implemented by the release code, and the parameters, including the spatial size, are tuned for best performance.

D. Performance Evaluation

In this section, three kinds of experiments are conducted. First, we randomly choose ten labeled samples per class for training and then compare the performance of all methods on each data set separately. Second, we verify the performance of all methods varied with different amount of training samples. Finally, we further test the performance of all methods influenced by noisy labels. To eliminate the influence of randomness in selecting labeled samples on the classification accuracy, all the numerical results in the following experiments are reported with the average results obtained from five rounds of different sampling.

1) Experimental Results With Fixed Amount of Training Samples: In this experiment, we compare the proposed method with other competing methods when ten training samples per class are labeled within the HSI. The numerical results on the Indian Pines, PaviaU, and KSC data sets are summarized in Tables III–V, respectively.

It can be seen that the proposed method produces pleasing classification results compared with other competing methods. Labeling only ten samples per class within the HSI, the proposed method surpasses the second-best method over 20 percent on all measures for the Indian Pine data set and PaviaU data set. By comparing the results from 3D-CNN, PRN, and SpecPatConvnet which directly trained via the labeled data, we can conclude the improvement of Ours over 3D-CNN, PRN, and SpecPatConvnet comes from the introduced unlabeled samples for training (i.e., the unlabeled data from the clustering task provides effective knowledge to constrain the overfitting problem the classification task encounters), even though PRN and SpecPatConvnet adopt advanced network structures, such as a residual network. Despite the success of the proposed method, the comparison experiments between Ours and the other semisupervised method (i.e., SS-LapSVM) indicate that the unlabeled data cannot always help to boost the classification accuracy when limited labeled training samples are provided. In addition, by comparing the results from Our and DCPN, which adopt a pair strategy to generate more labeled samples from the existing labeled data, we can see that the proposed method can steadily improve the classification accuracy on all data sets.

To better clarify the classification results, we visualize the obtained classification maps of different methods in Figs. 3–5 for each data set. We can find that the classification maps of the proposed method are the ones closest to the ground truth among all methods, which is consistent with the reported numerical results.

2) Experimental Results Varied With Different Amounts of Training Samples: In this experiment, we testify the performance of all methods varied with different amounts of training samples. Considering the problem, we focus on this study is limited labeled samples provided within the HSI for training, we utilize 5, 10, 15, 20, and 50 labeled samples per class for training, respectively. The numerical results on the Indian Pines, PaviaU, and KSC data sets are summarized in Tables VI–VIII, respectively. In addition, we plot the curves of all measures (i.e., OA, AA, and Kappa coefficient) versus the number of training samples for all methods in Figs. 6–8.

From these experimental results, we can see that the performance of all methods degrade when less labeled samples per class are adopted for training. Nevertheless, the proposed method has a better performance than any other competing

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TABLE IV

CLASSIFICATION RESULTS OF ALL METHODS PROVIDING TEN LABELED TRAINING SAMPLES PER CLASS ON PAVIAU DATA SET (%). BEST RESULTS ARE IN BOLD. THE NUMBERS IN PARENTHESES ARE STANDARD VARIANCES (%)

Class	SVM	3D-CNN	PRN	SpecPat Convnet	SS-LapSVM	DCPN	Ours
1	63.21	76.73	80.99	63.21	68.69	69.13	**99.52**
2	66.60	67.60	72.70	72.55	52.87	67.35	**98.12**
3	74.62	75.59	73.20	74.62	31.87	74.62	**100.00**
4	96.35	83.41	76.77	83.41	97.58	83.07	**100.00**
5	90.73	83.78	91.50	84.55	99.18	83.78	**99.22**
6	77.46	77.90	82.34	77.66	76.72	77.70	**100.00**
7	78.12	81.25	86.71	71.85	95.86	79.68	**96.87**
8	76.47	77.42	74.66	77.56	83.87	77.14	**99.44**
9	92.19	77.70	88.85	74.35	99.79	75.47	**100.00**
OA	$72.23_{(1.95)}$	$73.57_{(1.59)}$	$76.85_{(1.08)}$	$73.37_{(2.45)}$	$66.80_{(1.85)}$	$72.09_{(1.18)}$	$\mathbf{98.93}_{(0.56)}$
AA	$79.53_{(1.29)}$	$77.93_{(1.36)}$	$80.86_{(1.19)}$	$75.53_{(1.41)}$	$78.49_{(1.98)}$	$76.44_{(1.31)}$	$\mathbf{99.24}_{(0.20)}$
$\kappa \times 100$	$65.67_{(1.58)}$	$67.12_{(0.99)}$	$70.87_{(1.05)}$	$66.64_{(1.80)}$	$59.51_{(2.25)}$	$65.35_{(0.76)}$	$\mathbf{98.59}_{(0.72)}$

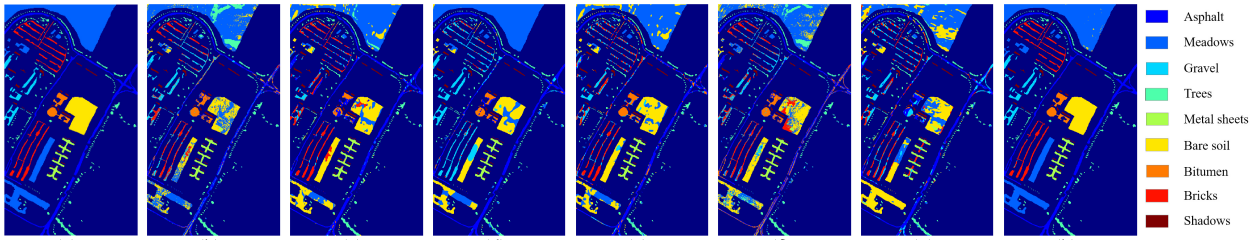


Fig. 4. Classification maps of different methods providing 10 labeled training samples per class on PaviaU data set. (a) Ground truth. (b) SVM. (c) 3D-CNN. (d) PRN. (e) SpecPatConvnet. (f) SS-LapSVM. (g) DCPN. (h) Ours.

TABLE V

CLASSIFICATION RESULTS OF ALL METHODS PROVIDING TEN LABELED TRAINING SAMPLES PER CLASS ON KSC DATA SET (%). BEST RESULTS ARE IN BOLD. THE NUMBERS IN PARENTHESES ARE STANDARD VARIANCES (%)

Class	SVM	3D-CNN	PRN	SpecPat Convnet	SS-LapSVM	DCPN	Ours
1	66.57	89.48	91.61	77.49	67.90	79.65	**97.60**
2	30.90	84.12	**98.28**	73.81	35.19	90.30	95.85
3	29.26	81.70	80.48	69.51	31.30	78.29	**98.54**
4	23.55	46.69	58.67	50.82	24.78	46.44	**97.52**
5	34.43	82.78	56.29	**82.11**	36.42	73.37	80.18
6	21.46	47.03	**92.69**	67.12	22.37	71.14	84.35
7	81.05	61.05	85.26	71.57	83.15	89.68	**98.18**
8	58.66	82.18	85.27	33.72	54.63	88.21	**96.86**
9	69.80	96.07	95.09	68.23	69.41	93.56	**97.29**
10	30.71	97.71	**99.74**	69.28	31.21	89.79	99.71
11	36.43	**100.00**	98.53	84.84	33.49	97.26	**100.00**
12	47.05	79.10	91.68	**97.56**	47.08	89.45	96.68
13	74.91	67.93	94.11	97.49	74.70	100.00	**100.00**
OA	$52.52_{(1.63)}$	$80.90_{(1.81)}$	$90.15_{(1.10)}$	$76.20_{(3.14)}$	$52.62_{(3.14)}$	$87.06_{(2.38)}$	$\mathbf{97.26}_{(1.92)}$
AA	$46.52_{(0.99)}$	$78.14_{(2.11)}$	$86.75_{(1.26)}$	$72.58_{(2.50)}$	$47.05_{(1.70)}$	$83.63_{(1.40)}$	$\mathbf{95.60}_{(1.23)}$
$\kappa \times 100$	$47.33_{(1.37)}$	$78.62_{(1.37)}$	$89.56_{(1.50)}$	$73.83_{(3.31)}$	$47.27_{(1.66)}$	$85.09_{(2.12)}$	$\mathbf{96.71}_{(0.97)}$

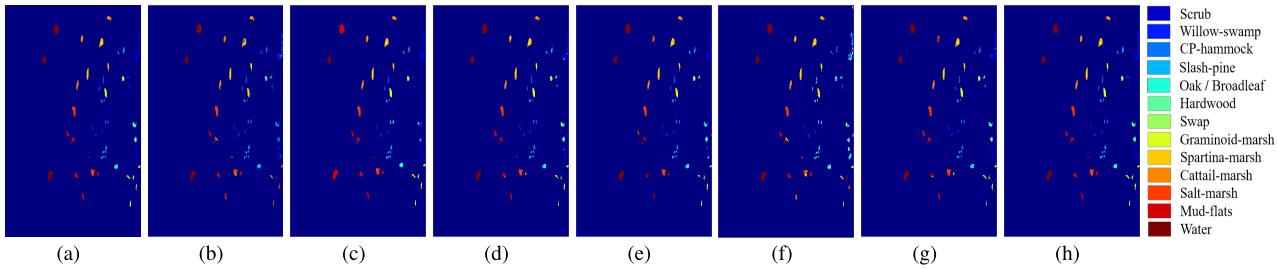


Fig. 5. Classification maps of different methods providing ten labeled training samples per class on KSC data set. (a) Ground truth. (b) SVM. (c) 3D-CNN. (d) PRN. (e) SpecPatConvnet. (f) SS-LapSVM. (g) DCPN. (h) Ours.

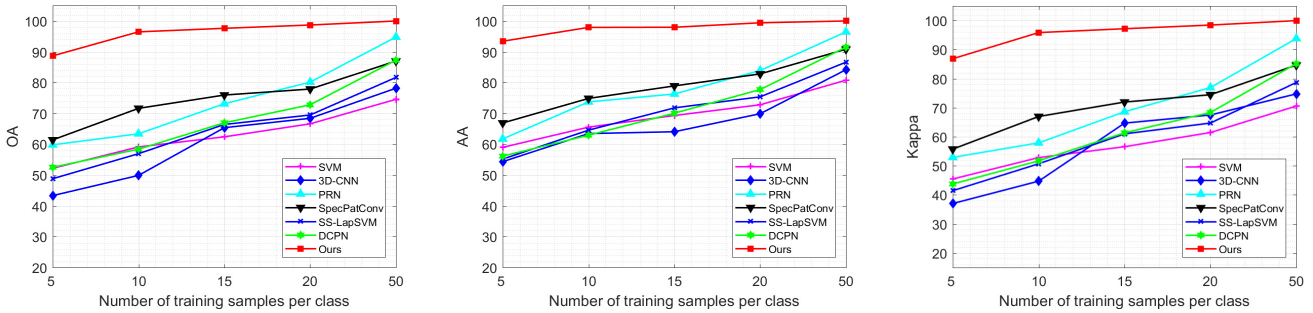
methods trained with any amount of labeled samples. More importantly, the proposed method has more obvious advantages over other competing methods with less labeled samples. When the labeled samples decrease to five samples per class, the proposed method improves OA over SpecPatConvnet more than 40%. Although SS-LapSVM can obtain better results over

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TABLE VI

CLASSIFICATION RESULTS OF ALL METHODS ON THE INDIAN PINES DATA SET VARIED WITH DIFFERENT AMOUNTS OF LABELED SAMPLES (%). BEST RESULTS ARE IN BOLD. THE NUMBERS IN PARENTHESES ARE STANDARD VARIANCES (%)

Number of data	Method	SVM	3D-CNN	PRN	SpecPat Convnet	SS-LapSVM	DCPN	Ours
5	OA	$52.18_{(1.95)}$	$43.31_{(3.07)}$	$59.79_{(2.31)}$	$61.40_{(2.52)}$	$48.80_{(2.87)}$	$52.51_{(2.48)}$	$\mathbf{88.67}_{(2.29)}$
	AA	$58.98_{(1.68)}$	$54.32_{(3.05)}$	$61.71_{(1.63)}$	$66.88_{(2.17)}$	$55.16_{(2.81)}$	$56.12_{(2.75)}$	$\mathbf{93.42}_{(1.78)}$
	$\kappa \times 100$	$45.46_{(2.03)}$	$37.02_{(2.16)}$	$52.91_{(1.80)}$	$55.74_{(2.32)}$	$41.44_{(2.67)}$	$43.73_{(1.40)}$	$\mathbf{86.86}_{(1.95)}$
10	OA	$59.30_{(1.16)}$	$50.63_{(3.26)}$	$62.71_{(1.39)}$	$71.70_{(2.39)}$	$57.84_{(1.73)}$	$58.29_{(2.15)}$	$\mathbf{96.30}_{(1.25)}$
	AA	$64.71_{(1.74)}$	$61.53_{(2.76)}$	$63.88_{(1.28)}$	$74.01_{(2.45)}$	$62.30_{(2.64)}$	$62.25_{(3.04)}$	$\mathbf{97.93}_{(1.16)}$
	$\kappa \times 100$	$53.85_{(1.25)}$	$44.78_{(2.28)}$	$57.54_{(1.29)}$	$67.87_{(1.83)}$	$52.67_{(2.25)}$	$53.13_{(2.15)}$	$\mathbf{95.68}_{(1.86)}$
15	OA	$62.39_{(1.24)}$	$65.36_{(2.63)}$	$73.12_{(1.46)}$	$75.95_{(1.39)}$	$66.40_{(1.76)}$	$66.94_{(2.28)}$	$\mathbf{97.61}_{(1.00)}$
	AA	$69.31_{(1.11)}$	$64.04_{(2.66)}$	$76.30_{(1.72)}$	$78.91_{(1.45)}$	$71.78_{(1.38)}$	$70.05_{(2.14)}$	$\mathbf{97.95}_{(1.14)}$
	$\kappa \times 100$	$56.58_{(1.14)}$	$64.70_{(2.53)}$	$68.61_{(1.15)}$	$71.95_{(2.13)}$	$61.00_{(1.27)}$	$61.36_{(1.50)}$	$\mathbf{97.19}_{(1.57)}$
20	OA	$66.61_{(1.25)}$	$68.40_{(1.28)}$	$80.15_{(1.24)}$	$77.90_{(1.11)}$	$69.48_{(1.35)}$	$72.79_{(2.47)}$	$\mathbf{98.68}_{(0.57)}$
	AA	$72.79_{(1.03)}$	$69.89_{(2.81)}$	$83.91_{(0.95)}$	$82.81_{(1.76)}$	$75.37_{(1.15)}$	$77.74_{(1.83)}$	$\mathbf{99.40}_{(0.36)}$
	$\kappa \times 100$	$61.46_{(1.85)}$	$67.49_{(1.76)}$	$76.93_{(1.35)}$	$74.47_{(1.17)}$	$64.72_{(1.06)}$	$68.36_{(1.89)}$	$\mathbf{98.45}_{(0.28)}$
50	OA	$74.56_{(0.88)}$	$79.17_{(1.35)}$	$94.88_{(1.25)}$	$86.85_{(1.46)}$	$81.67_{(0.84)}$	$87.27_{(2.44)}$	$\mathbf{99.98}_{(0.01)}$
	AA	$80.77_{(1.02)}$	$84.17_{(1.48)}$	$96.38_{(0.69)}$	$90.82_{(0.96)}$	$86.62_{(0.37)}$	$91.49_{(2.31)}$	$\mathbf{99.99}_{(0.01)}$
	$\kappa \times 100$	$70.57_{(0.94)}$	$74.76_{(1.24)}$	$94.03_{(1.13)}$	$84.66_{(1.59)}$	$78.68_{(0.92)}$	$85.16_{(1.73)}$	$\mathbf{99.98}_{(0.01)}$



(a) (b) (c)

Fig. 6. Classification results of all methods on the Indian pines data set varied with different amounts of labeled samples. (a) OA. (b) AA. (c) $\kappa \times 100$.

TABLE VII

CLASSIFICATION RESULTS OF ALL METHODS ON THE PAVIAU DATA SET VARIED WITH DIFFERENT AMOUNTS OF LABELED SAMPLES (%). BEST RESULTS ARE IN BOLD. THE NUMBERS IN PARENTHESES ARE STANDARD VARIANCES (%)

Number of data	Method	SVM	3D-CNN	PRN	SpecPat Convnet	SS-LapSVM	DCPN	Ours
5	OA	$66.95_{(2.75)}$	$63.72_{(1.15)}$	$68.98_{(1.69)}$	$53.81_{(2.40)}$	$55.53_{(3.92)}$	$66.12_{(1.93)}$	$\mathbf{95.21}_{(1.76)}$
	AA	$73.48_{(3.59)}$	$68.57_{(2.14)}$	$68.78_{(1.44)}$	$70.93_{(2.47)}$	$70.27_{(2.14)}$	$73.40_{(1.52)}$	$\mathbf{98.49}_{(1.80)}$
	$\kappa \times 100$	$58.05_{(2.52)}$	$53.14_{(1.48)}$	$60.95_{(1.35)}$	$45.16_{(2.48)}$	$46.32_{(1.81)}$	$55.86_{(1.40)}$	$\mathbf{93.61}_{(1.19)}$
10	OA	$72.23_{(1.95)}$	$73.57_{(1.59)}$	$76.85_{(1.08)}$	$73.37_{(2.45)}$	$66.80_{(1.85)}$	$72.09_{(1.18)}$	$\mathbf{98.93}_{(0.56)}$
	AA	$79.53_{(1.29)}$	$77.93_{(1.36)}$	$80.86_{(1.19)}$	$75.53_{(1.41)}$	$78.49_{(1.98)}$	$76.44_{(1.31)}$	$\mathbf{99.24}_{(0.20)}$
	$\kappa \times 100$	$65.67_{(1.58)}$	$67.12_{(0.99)}$	$70.87_{(1.05)}$	$66.64_{(1.80)}$	$59.51_{(2.25)}$	$65.35_{(0.76)}$	$\mathbf{98.59}_{(0.72)}$
15	OA	$73.49_{(1.94)}$	$75.14_{(0.90)}$	$81.05_{(1.40)}$	$75.32_{(1.92)}$	$77.41_{(1.91)}$	$79.79_{(1.32)}$	$\mathbf{99.25}_{(0.17)}$
	AA	$82.20_{(2.19)}$	$82.60_{(0.76)}$	$82.78_{(1.47)}$	$80.89_{(1.76)}$	$85.83_{(1.74)}$	$83.04_{(1.23)}$	$\mathbf{99.79}_{(0.12)}$
	$\kappa \times 100$	$66.47_{(1.90)}$	$68.95_{(1.56)}$	$75.31_{(1.48)}$	$68.20_{(2.00)}$	$71.63_{(2.04)}$	$73.60_{(0.92)}$	$\mathbf{99.01}_{(0.25)}$
20	OA	$77.24_{(1.15)}$	$76.11_{(0.82)}$	$83.55_{(0.41)}$	$80.57_{(1.48)}$	$79.13_{(1.20)}$	$83.51_{(0.65)}$	$\mathbf{99.79}_{(0.11)}$
	AA	$83.30_{(1.14)}$	$84.15_{(1.48)}$	$86.33_{(0.80)}$	$88.64_{(1.72)}$	$86.93_{(1.90)}$	$85.73_{(0.77)}$	$\mathbf{99.92}_{(0.06)}$
	$\kappa \times 100$	$70.70_{(0.48)}$	$69.73_{(0.43)}$	$79.37_{(0.92)}$	$75.38_{(1.55)}$	$73.64_{(1.12)}$	$78.41_{(0.81)}$	$\mathbf{99.72}_{(0.24)}$
50	OA	$84.35_{(0.59)}$	$90.66_{(0.79)}$	$94.34_{(0.76)}$	$91.63_{(0.50)}$	$89.90_{(0.90)}$	$92.78_{(0.31)}$	$\mathbf{99.99}_{(0.00)}$
	AA	$87.83_{(0.36)}$	$94.18_{(0.64)}$	$94.97_{(0.86)}$	$94.29_{(1.14)}$	$92.47_{(1.27)}$	$94.16_{(0.83)}$	$\mathbf{99.99}_{(0.01)}$
	$\kappa \times 100$	$79.68_{(0.49)}$	$87.80_{(0.69)}$	$92.52_{(0.46)}$	$89.10_{(0.81)}$	$86.86_{(0.85)}$	$90.43_{(0.24)}$	$\mathbf{99.99}_{(0.01)}$

its counterpart supervised method (i.e., SVM) providing 15 or 20 labeled samples per class for training, it can be seen that the classification accuracy of SS-LapSVM is lower than SVM when only five or ten labeled samples per class are provided, which indicates that the difficulty of utilizing the unlabeled data for HSI classification is enhanced a lot with a few labeled training samples. Nevertheless, the proposed method Ours can effectively utilize the unlabeled data for classification within the proposed framework.

3) Experimental Results With Noisy Labels: In this experiment, we testify the performance of all methods influenced by label noise when 20 labeled samples per class are provided for training. Specifically, we randomly replace part of the labels of the training samples with incorrect labels, among which the ratio of replacement is varied from 0.05 to 0.5. The numerical results on the Indian Pines, PaviaU, and KSC data sets are summarized in Tables IX–XI, respectively. In addition, we plot the curves of all measures versus the

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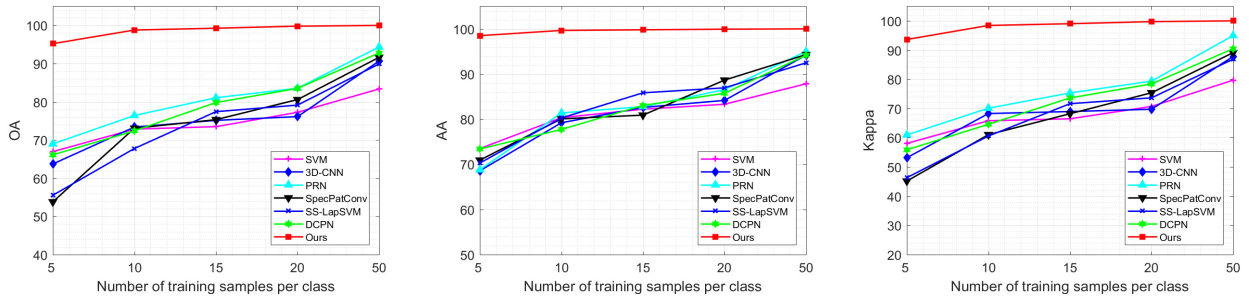


Fig. 7. Classification results of all methods on the PaviaU data set varied with different amounts of labeled samples. (a) OA. (b) AA. (c) $\kappa \times 100$.

TABLE VIII

CLASSIFICATION RESULTS OF ALL METHODS ON THE KSC DATA SET VARIED WITH DIFFERENT AMOUNTS OF LABELED SAMPLES (%). BEST RESULTS ARE IN BOLD. THE NUMBERS IN PARENTHESES ARE STANDARD VARIANCES (%)

Number of data	Method	SVM	3D-CNN	PRN	SpecPat Convnet	SS-LapSVM	DCPN	Ours
5	OA	$51.99_{(0.46)}$	$73.59_{(3.80)}$	$85.68_{(1.53)}$	$58.44_{(3.48)}$	$44.56_{(1.50)}$	$81.14_{(3.50)}$	$\mathbf{95.36}_{(1.66)}$
	AA	$46.50_{(0.87)}$	$66.80_{(2.46)}$	$80.03_{(2.12)}$	$52.86_{(3.25)}$	$34.58_{(1.84)}$	$75.25_{(2.57)}$	$\mathbf{93.45}_{(1.88)}$
	$\kappa \times 100$	$46.62_{(1.06)}$	$70.65_{(2.98)}$	$84.03_{(2.03)}$	$55.50_{(4.01)}$	$38.58_{(2.03)}$	$78.99_{(3.06)}$	$\mathbf{94.71}_{(2.13)}$
10	OA	$52.52_{(1.63)}$	$80.90_{(1.81)}$	$90.15_{(1.10)}$	$76.20_{(3.14)}$	$52.62_{(2.07)}$	$87.06_{(2.38)}$	$\mathbf{97.26}_{(1.92)}$
	AA	$46.52_{(0.99)}$	$78.14_{(2.11)}$	$86.75_{(1.26)}$	$72.58_{(2.50)}$	$47.05_{(1.70)}$	$83.63_{(1.40)}$	$\mathbf{95.60}_{(1.23)}$
	$\kappa \times 100$	$47.33_{(1.37)}$	$78.62_{(1.37)}$	$89.56_{(1.50)}$	$73.83_{(3.31)}$	$47.27_{(1.66)}$	$85.09_{(2.12)}$	$\mathbf{96.71}_{(0.97)}$
15	OA	$55.21_{(1.60)}$	$87.63_{(1.67)}$	$96.13_{(1.24)}$	$88.14_{(2.71)}$	$61.89_{(1.76)}$	$90.23_{(1.92)}$	$\mathbf{98.53}_{(0.74)}$
	AA	$47.90_{(1.05)}$	$82.36_{(1.30)}$	$93.76_{(0.96)}$	$84.87_{(2.38)}$	$48.06_{(1.66)}$	$86.71_{(2.40)}$	$\mathbf{98.20}_{(0.73)}$
	$\kappa \times 100$	$49.34_{(0.75)}$	$86.22_{(1.98)}$	$95.68_{(1.40)}$	$87.11_{(2.40)}$	$57.36_{(0.92)}$	$89.10_{(1.78)}$	$\mathbf{98.44}_{(0.58)}$
20	OA	$62.29_{(1.67)}$	$88.64_{(2.12)}$	$97.96_{(1.76)}$	$90.88_{(2.60)}$	$67.70_{(1.26)}$	$94.14_{(1.59)}$	$\mathbf{98.77}_{(0.51)}$
	AA	$55.22_{(0.92)}$	$84.32_{(1.13)}$	$96.89_{(0.78)}$	$88.09_{(2.43)}$	$54.45_{(2.39)}$	$91.04_{(1.46)}$	$\mathbf{98.45}_{(0.58)}$
	$\kappa \times 100$	$58.07_{(1.55)}$	$87.33_{(1.26)}$	$97.72_{(1.17)}$	$89.83_{(1.95)}$	$63.79_{(1.75)}$	$93.45_{(2.50)}$	$\mathbf{98.64}_{(0.59)}$
50	OA	$75.59_{(1.65)}$	$95.95_{(1.74)}$	$99.52_{(0.24)}$	$93.61_{(2.46)}$	$84.51_{(1.31)}$	$97.82_{(0.36)}$	$\mathbf{99.84}_{(0.04)}$
	AA	$77.39_{(1.80)}$	$93.79_{(1.51)}$	$99.16_{(0.17)}$	$90.69_{(1.96)}$	$80.92_{(2.16)}$	$97.03_{(1.40)}$	$\mathbf{99.86}_{(0.02)}$
	$\kappa \times 100$	$73.00_{(1.53)}$	$95.45_{(1.89)}$	$99.47_{(0.34)}$	$92.83_{(1.82)}$	$82.64_{(1.36)}$	$97.56_{(1.41)}$	$\mathbf{99.82}_{(0.02)}$

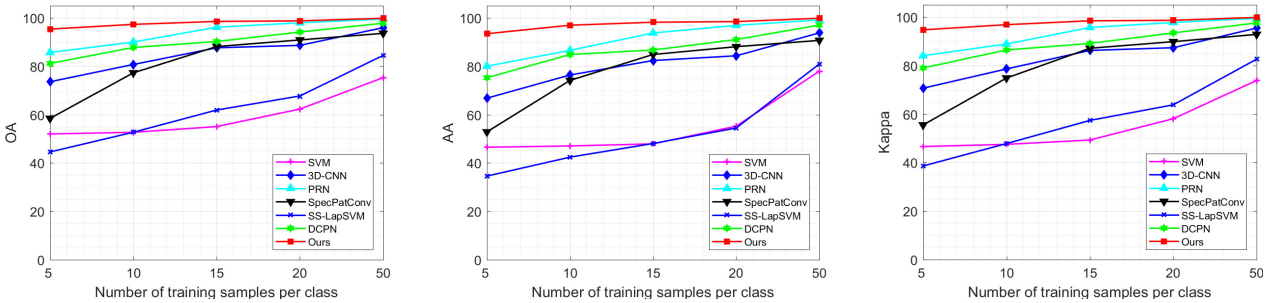


Fig. 8. Classification results of all methods on the KSC data set varied with different amounts of labeled samples. (a) OA. (b) AA. (c) $\kappa \times 100$.

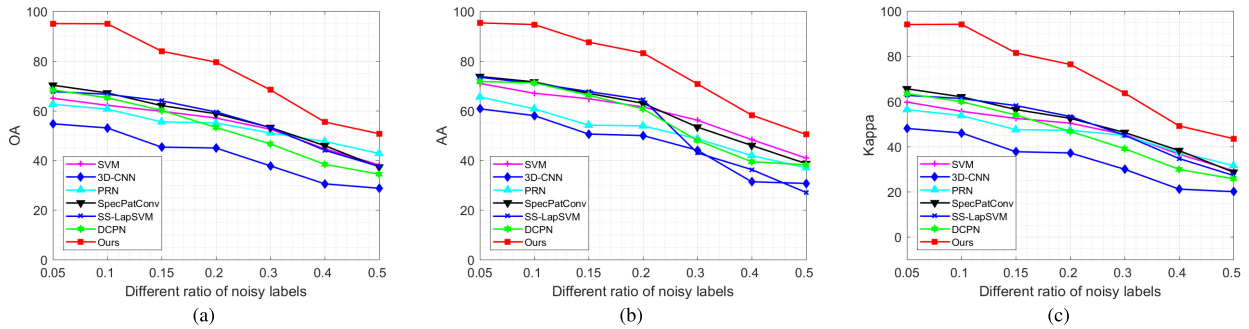


Fig. 9. Classification results of all methods on the Indian Pines data set varied with different ratio of noisy labels. (a) OA. (b) AA. (c) $\kappa \times 100$.

ratio of noisy labels for all methods in Figs. 9–11. It can be seen that the proposed method obtains the best performance over other competing methods with any ratio of noisy labels.

In summary, from the above-mentioned experiments on both accuracy and robustness to noise, we can conclude that the proposed method has obvious advantages over other state-of-the-art methods.

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TABLE IX

CLASSIFICATION RESULTS OF ALL METHODS ON THE INDIAN PINES DATA SET VARIED WITH DIFFERENT RATIO OF NOISY LABELS. (%). BEST RESULTS ARE IN BOLD. THE NUMBERS IN PARENTHESES ARE STANDARD VARIANCES (%)

Ratio of Noisy Labels	Method	SVM	3D-CNN	PRN	SpecPat Convnet	SS-LapSVM	DCPN	Ours
0.05	OA	$64.79_{(1.92)}$	$54.67_{(2.19)}$	$62.55_{(2.05)}$	$70.22_{(2.61)}$	$67.66_{(1.97)}$	$68.31_{(2.32)}$	$\mathbf{95.03}_{(1.98)}$
	AA	$70.93_{(2.13)}$	$60.72_{(2.12)}$	$65.49_{(1.80)}$	$73.72_{(2.29)}$	$73.40_{(1.89)}$	$71.75_{(2.21)}$	$\mathbf{95.32}_{(1.93)}$
	$\kappa \times 100$	$59.68_{(2.50)}$	$48.02_{(1.97)}$	$56.35_{(2.12)}$	$65.55_{(2.71)}$	$62.66_{(2.08)}$	$63.42_{(2.83)}$	$\mathbf{94.00}_{(1.86)}$
0.1	OA	$62.12_{(2.40)}$	$53.02_{(2.03)}$	$60.60_{(2.03)}$	$67.11_{(2.86)}$	$66.58_{(2.25)}$	$65.17_{(2.39)}$	$\mathbf{94.97}_{(2.42)}$
	AA	$66.97_{(2.21)}$	$57.99_{(3.05)}$	$60.70_{(2.78)}$	$71.56_{(2.16)}$	$71.00_{(2.89)}$	$71.16_{(2.71)}$	$\mathbf{94.65}_{(1.98)}$
	$\kappa \times 100$	$55.59_{(2.43)}$	$46.04_{(2.71)}$	$53.71_{(2.71)}$	$62.01_{(1.86)}$	$61.24_{(2.78)}$	$59.87_{(2.17)}$	$\mathbf{94.10}_{(2.27)}$
0.15	OA	$59.72_{(2.60)}$	$45.28_{(1.17)}$	$55.47_{(2.97)}$	$62.01_{(2.16)}$	$63.94_{(2.08)}$	$60.10_{(2.02)}$	$\mathbf{83.91}_{(2.70)}$
	AA	$64.78_{(3.03)}$	$50.56_{(1.37)}$	$54.15_{(2.17)}$	$66.95_{(2.95)}$	$67.76_{(2.36)}$	$66.21_{(2.54)}$	$\mathbf{87.57}_{(2.86)}$
	$\kappa \times 100$	$52.47_{(2.28)}$	$37.71_{(2.80)}$	$47.46_{(1.41)}$	$56.38_{(3.17)}$	$58.14_{(1.96)}$	$53.95_{(1.98)}$	$\mathbf{81.45}_{(2.74)}$
0.2	OA	$57.00_{(2.26)}$	$44.94_{(2.74)}$	$55.03_{(2.41)}$	$58.74_{(1.50)}$	$59.46_{(2.13)}$	$53.21_{(2.55)}$	$\mathbf{79.49}_{(1.61)}$
	AA	$61.45_{(1.05)}$	$49.94_{(2.62)}$	$53.84_{(2.97)}$	$62.94_{(1.71)}$	$64.42_{(2.03)}$	$60.69_{(1.90)}$	$\mathbf{83.20}_{(1.50)}$
	$\kappa \times 100$	$50.34_{(2.31)}$	$37.18_{(2.17)}$	$47.20_{(2.99)}$	$52.49_{(2.62)}$	$53.32_{(1.85)}$	$46.73_{(2.37)}$	$\mathbf{76.33}_{(2.02)}$
0.3	OA	$52.50_{(2.48)}$	$37.75_{(2.31)}$	$50.99_{(2.57)}$	$53.19_{(2.33)}$	$53.12_{(3.14)}$	$46.56_{(2.76)}$	$\mathbf{68.35}_{(3.06)}$
	AA	$56.14_{(2.96)}$	$44.07_{(1.95)}$	$48.64_{(3.79)}$	$53.37_{(2.74)}$	$42.96_{(2.33)}$	$47.99_{(2.51)}$	$\mathbf{70.74}_{(2.97)}$
	$\kappa \times 100$	$45.32_{(1.69)}$	$29.93_{(2.55)}$	$44.92_{(2.03)}$	$46.22_{(2.77)}$	$45.02_{(3.41)}$	$39.05_{(2.96)}$	$\mathbf{63.67}_{(2.68)}$
0.4	OA	$44.64_{(1.20)}$	$30.50_{(1.17)}$	$47.58_{(2.97)}$	$45.89_{(1.71)}$	$44.04_{(2.42)}$	$38.36_{(1.90)}$	$\mathbf{55.42}_{(3.55)}$
	AA	$48.30_{(2.17)}$	$31.35_{(2.92)}$	$41.89_{(1.00)}$	$45.96_{(2.98)}$	$36.24_{(2.34)}$	$39.40_{(2.27)}$	$\mathbf{58.18}_{(3.29)}$
	$\kappa \times 100$	$36.61_{(1.34)}$	$21.17_{(2.59)}$	$37.44_{(1.90)}$	$38.23_{(1.09)}$	$34.70_{(3.70)}$	$29.90_{(2.18)}$	$\mathbf{49.14}_{(3.36)}$
0.5	OA	$38.11_{(2.94)}$	$28.72_{(1.23)}$	$42.75_{(1.76)}$	$37.29_{(1.57)}$	$37.53_{(2.65)}$	$34.39_{(2.38)}$	$\mathbf{50.69}_{(3.67)}$
	AA	$40.89_{(2.03)}$	$30.66_{(2.89)}$	$36.98_{(2.39)}$	$38.64_{(2.86)}$	$27.02_{(2.92)}$	$38.18_{(2.37)}$	$\mathbf{50.40}_{(2.72)}$
	$\kappa \times 100$	$29.32_{(1.08)}$	$20.03_{(1.38)}$	$31.59_{(2.87)}$	$28.64_{(1.16)}$	$27.19_{(2.11)}$	$25.73_{(2.77)}$	$\mathbf{43.49}_{(3.79)}$

TABLE X

CLASSIFICATION RESULTS OF ALL METHODS ON PAVIAU DATA SET VARIED WITH DIFFERENT RATIO OF NOISY LABELS (%) BEST RESULTS ARE IN BOLD. THE NUMBERS IN PARENTHESES ARE STANDARD VARIANCES (%)

Ratio of Noisy Labels	Method	SVM	3D-CNN	PRN	SpecPat Convnet	SS-LapSVM	DCPN	Ours
0.05	OA	$74.28_{(1.14)}$	$73.08_{(1.74)}$	$74.82_{(1.11)}$	$75.99_{(1.30)}$	$75.23_{(1.30)}$	$81.79_{(1.70)}$	$\mathbf{97.94}_{(1.42)}$
	AA	$79.37_{(1.18)}$	$77.43_{(1.36)}$	$70.41_{(1.37)}$	$82.34_{(1.07)}$	$83.95_{(1.07)}$	$81.86_{(1.99)}$	$\mathbf{98.21}_{(1.31)}$
	$\kappa \times 100$	$67.43_{(1.06)}$	$65.73_{(1.64)}$	$65.90_{(1.34)}$	$69.01_{(2.12)}$	$68.84_{(1.61)}$	$76.00_{(0.54)}$	$\mathbf{97.28}_{(0.93)}$
0.1	OA	$70.95_{(1.69)}$	$62.34_{(2.48)}$	$73.12_{(1.73)}$	$75.37_{(1.25)}$	$74.22_{(2.11)}$	$79.42_{(1.67)}$	$\mathbf{95.59}_{(2.17)}$
	AA	$79.76_{(2.20)}$	$67.73_{(1.64)}$	$66.50_{(1.84)}$	$81.95_{(1.61)}$	$80.67_{(1.41)}$	$80.49_{(1.76)}$	$\mathbf{96.69}_{(1.77)}$
	$\kappa \times 100$	$63.74_{(1.63)}$	$53.14_{(2.36)}$	$63.28_{(2.03)}$	$69.12_{(2.41)}$	$67.65_{(1.84)}$	$73.03_{(1.75)}$	$\mathbf{94.30}_{(2.07)}$
0.15	OA	$69.68_{(2.37)}$	$57.89_{(2.60)}$	$71.30_{(1.74)}$	$73.04_{(2.84)}$	$66.82_{(2.65)}$	$74.15_{(2.75)}$	$\mathbf{91.04}_{(2.35)}$
	AA	$75.83_{(2.53)}$	$61.99_{(1.41)}$	$66.75_{(2.52)}$	$77.74_{(2.65)}$	$76.18_{(2.78)}$	$77.82_{(1.95)}$	$\mathbf{92.54}_{(1.05)}$
	$\kappa \times 100$	$61.95_{(2.66)}$	$48.16_{(2.17)}$	$61.71_{(2.44)}$	$65.08_{(2.78)}$	$59.34_{(2.09)}$	$66.80_{(2.92)}$	$\mathbf{88.73}_{(2.00)}$
0.2	OA	$69.49_{(1.21)}$	$52.00_{(2.25)}$	$69.34_{(3.11)}$	$65.02_{(1.41)}$	$65.12_{(3.41)}$	$70.92_{(1.76)}$	$\mathbf{83.56}_{(2.88)}$
	AA	$73.75_{(2.59)}$	$57.76_{(1.65)}$	$65.31_{(2.65)}$	$71.26_{(2.09)}$	$75.80_{(2.76)}$	$75.42_{(2.07)}$	$\mathbf{87.79}_{(1.31)}$
	$\kappa \times 100$	$61.26_{(2.43)}$	$42.10_{(3.45)}$	$59.00_{(2.36)}$	$56.85_{(2.88)}$	$58.66_{(3.86)}$	$63.12_{(2.89)}$	$\mathbf{79.59}_{(2.91)}$
0.3	OA	$67.13_{(2.39)}$	$50.13_{(1.92)}$	$66.97_{(3.23)}$	$60.78_{(2.89)}$	$63.03_{(1.28)}$	$69.34_{(2.10)}$	$\mathbf{75.76}_{(1.36)}$
	AA	$72.13_{(1.44)}$	$48.48_{(2.04)}$	$59.86_{(1.41)}$	$59.61_{(2.56)}$	$56.43_{(3.53)}$	$67.21_{(3.35)}$	$\mathbf{75.99}_{(1.71)}$
	$\kappa \times 100$	$59.39_{(2.92)}$	$40.12_{(1.80)}$	$57.94_{(3.47)}$	$52.01_{(2.42)}$	$53.35_{(2.00)}$	$61.63_{(2.56)}$	$\mathbf{71.89}_{(1.58)}$
0.4	OA	$61.11_{(2.32)}$	$41.03_{(2.04)}$	$58.17_{(3.22)}$	$51.38_{(2.47)}$	$56.30_{(2.01)}$	$63.07_{(2.65)}$	$\mathbf{68.91}_{(2.53)}$
	AA	$51.83_{(2.92)}$	$34.26_{(1.85)}$	$46.87_{(3.57)}$	$48.50_{(2.60)}$	$46.08_{(3.80)}$	$62.34_{(2.73)}$	$\mathbf{69.91}_{(1.60)}$
	$\kappa \times 100$	$51.72_{(1.48)}$	$28.66_{(2.05)}$	$47.32_{(2.09)}$	$41.29_{(2.66)}$	$45.56_{(2.67)}$	$54.43_{(1.43)}$	$\mathbf{64.16}_{(1.79)}$
0.5	OA	$57.92_{(1.86)}$	$33.58_{(2.67)}$	$56.51_{(2.65)}$	$42.79_{(2.68)}$	$53.00_{(2.85)}$	$57.21_{(1.17)}$	$\mathbf{64.62}_{(2.33)}$
	AA	$46.51_{(1.01)}$	$31.67_{(1.96)}$	$45.25_{(2.21)}$	$42.33_{(2.26)}$	$41.60_{(2.57)}$	$54.52_{(2.31)}$	$\mathbf{62.12}_{(1.58)}$
	$\kappa \times 100$	$47.59_{(1.29)}$	$22.15_{(2.29)}$	$45.79_{(2.74)}$	$32.04_{(2.55)}$	$41.44_{(2.13)}$	$47.20_{(1.88)}$	$\mathbf{56.42}_{(2.83)}$

E. Ablation Study

In this section, to further testify the proposed method, we investigate the effect of different feature fusion module, the effect of different weights for the constructed loss, the effect of the clustering number, and the effect of cube size for classification in this section. The experiments are conducted on the Indian Pines data set when 20 labeled samples per class are provided for training.

1) Effect of Different Feature Fusion Strategies: As we introduced in the Section II-D of the proposed method, there are six strategies for feature fusion. In this experiment, we testify the performance of these fusion strategies in terms of HSI classification and summarize the numerical results in Table XII. It can be seen that the variance of different feature fusion strategies is within a small range, i.e., bilinear and concatenation have slight advantages over other feature fusion strategies in terms of OA, AA, and Kappa measures. From the experiment, we can see that the proposed method is rather stable with different feature fusion strategies. In this study, we choose bilinear for feature fusion for all experiments.

2) Effect of Different Weights for the Constructed Loss: The constructed loss function of the proposed method consists of three parts, as shown in Formula (13). Different weight combinations will result in different classification results. To verify the performance of the proposed method varied with different

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TABLE XI

CLASSIFICATION RESULTS OF ALL METHODS ON KSC DATA SET VARIED WITH DIFFERENT RATIO OF NOISY LABELS (%). BEST RESULTS ARE IN BOLD. THE NUMBERS IN PARENTHESES ARE STANDARD VARIANCES (%)

Ratio of Noisy Labels	Method	SVM	3D-CNN	PRN	SpecPat Convnet	SS-LapSVM	DCPN	Ours
0.05	OA	$46.87_{(1.29)}$	$82.91_{(1.52)}$	$88.19_{(1.74)}$	$81.97_{(2.09)}$	$62.97_{(1.28)}$	$90.46_{(1.68)}$	$\mathbf{95.31}_{(1.73)}$
	AA	$37.55_{(1.65)}$	$78.56_{(0.74)}$	$84.39_{(1.37)}$	$77.88_{(1.78)}$	$51.69_{(1.51)}$	$88.07_{(1.69)}$	$\mathbf{94.61}_{(1.10)}$
	$\kappa \times 100$	$40.95_{(2.07)}$	$80.96_{(1.55)}$	$86.81_{(1.61)}$	$79.94_{(1.98)}$	$58.85_{(1.62)}$	$89.37_{(1.41)}$	$\mathbf{94.77}_{(1.86)}$
0.1	OA	$44.13_{(2.00)}$	$76.88_{(1.45)}$	$84.28_{(2.11)}$	$80.59_{(2.88)}$	$59.31_{(2.78)}$	$89.49_{(2.38)}$	$\mathbf{93.31}_{(1.77)}$
	AA	$36.85_{(2.25)}$	$70.90_{(2.52)}$	$79.83_{(1.38)}$	$79.10_{(1.49)}$	$51.49_{(1.98)}$	$86.64_{(0.95)}$	$\mathbf{90.90}_{(2.48)}$
	$\kappa \times 100$	$40.00_{(1.62)}$	$74.21_{(2.44)}$	$82.59_{(2.11)}$	$78.50_{(1.82)}$	$55.06_{(1.95)}$	$88.27_{(1.98)}$	$\mathbf{92.53}_{(1.34)}$
0.15	OA	$41.42_{(2.15)}$	$74.45_{(2.78)}$	$79.32_{(2.48)}$	$79.91_{(2.84)}$	$50.46_{(2.48)}$	$86.66_{(3.95)}$	$\mathbf{88.67}_{(2.48)}$
	AA	$34.21_{(2.97)}$	$67.96_{(2.00)}$	$73.10_{(2.11)}$	$76.35_{(2.54)}$	$43.51_{(2.95)}$	$82.85_{(3.52)}$	$\mathbf{86.88}_{(2.77)}$
	$\kappa \times 100$	$35.25_{(2.87)}$	$71.56_{(2.44)}$	$76.90_{(2.14)}$	$78.76_{(1.63)}$	$45.30_{(1.19)}$	$85.17_{(3.55)}$	$\mathbf{87.38}_{(1.98)}$
0.2	OA	$41.12_{(1.01)}$	$66.82_{(1.78)}$	$77.25_{(2.37)}$	$75.08_{(1.32)}$	$48.88_{(2.52)}$	$80.06_{(3.19)}$	$\mathbf{84.17}_{(2.34)}$
	AA	$33.65_{(2.93)}$	$59.76_{(2.26)}$	$72.67_{(2.01)}$	$71.09_{(1.99)}$	$40.71_{(2.92)}$	$77.65_{(2.52)}$	$\mathbf{82.56}_{(2.39)}$
	$\kappa \times 100$	$35.10_{(2.44)}$	$63.15_{(2.79)}$	$74.82_{(1.59)}$	$72.33_{(2.02)}$	$43.48_{(1.91)}$	$77.85_{(3.20)}$	$\mathbf{82.40}_{(2.27)}$
0.3	OA	$40.12_{(2.44)}$	$63.83_{(1.40)}$	$75.21_{(1.24)}$	$73.28_{(2.04)}$	$48.04_{(1.49)}$	$76.08_{(3.12)}$	$\mathbf{78.32}_{(2.06)}$
	AA	$33.18_{(1.93)}$	$53.40_{(2.29)}$	$71.50_{(2.12)}$	$61.56_{(1.38)}$	$34.20_{(1.81)}$	$64.70_{(2.10)}$	$\mathbf{72.99}_{(3.74)}$
	$\kappa \times 100$	$33.99_{(1.57)}$	$59.70_{(1.74)}$	$72.56_{(2.39)}$	$70.16_{(1.65)}$	$42.14_{(2.81)}$	$73.38_{(3.41)}$	$\mathbf{76.20}_{(2.36)}$
0.4	OA	$35.97_{(2.21)}$	$59.99_{(1.42)}$	$67.75_{(2.13)}$	$65.83_{(2.57)}$	$46.41_{(1.60)}$	$68.12_{(3.02)}$	$\mathbf{69.63}_{(2.87)}$
	AA	$27.92_{(2.95)}$	$50.23_{(2.49)}$	$67.20_{(1.82)}$	$55.43_{(1.81)}$	$33.07_{(2.10)}$	$61.12_{(2.91)}$	$\mathbf{62.99}_{(2.30)}$
	$\kappa \times 100$	$29.28_{(1.22)}$	$55.47_{(2.71)}$	$64.45_{(2.15)}$	$61.95_{(1.60)}$	$40.46_{(2.62)}$	$64.82_{(4.43)}$	$\mathbf{66.38}_{(1.31)}$
0.5	OA	$35.89_{(1.47)}$	$47.00_{(1.74)}$	$59.37_{(2.24)}$	$51.49_{(1.91)}$	$42.21_{(1.78)}$	$57.69_{(3.30)}$	$\mathbf{60.17}_{(2.15)}$
	AA	$27.56_{(2.52)}$	$39.05_{(2.00)}$	$56.43_{(3.11)}$	$44.90_{(2.26)}$	$29.09_{(2.38)}$	$48.30_{(2.54)}$	$\mathbf{57.21}_{(3.05)}$
	$\kappa \times 100$	$29.22_{(2.50)}$	$41.34_{(1.80)}$	$54.61_{(2.07)}$	$46.44_{(1.85)}$	$35.81_{(1.98)}$	$52.95_{(3.32)}$	$\mathbf{55.98}_{(3.07)}$

TABLE XII

EFFECT OF DIFFERENT FEATURE FUSION STRATEGIES FOR HSI CLASSIFICATION (%). BEST RESULTS ARE IN BOLD. THE NUMBERS IN PARENTHESES ARE STANDARD VARIANCES (%)

Fusion Method	OA	AA	$\kappa \times 100$
Concatentation	$\mathbf{98.85}_{(0.78)}$	$99.26_{(0.28)}$	$\mathbf{98.65}_{(0.65)}$
Attention	$97.82_{(1.26)}$	$98.61_{(1.03)}$	$97.53_{(1.32)}$
Transformation	$97.88_{(1.60)}$	$98.72_{(1.13)}$	$97.46_{(1.64)}$
Multiplication	$98.02_{(1.29)}$	$98.96_{(0.47)}$	$97.68_{(1.38)}$
Convolution	$98.30_{(0.52)}$	$99.11_{(0.59)}$	$98.01_{(1.53)}$
Bilinear	$98.68_{(0.57)}$	$\mathbf{99.40}_{(0.36)}$	$98.45_{(0.28)}$

TABLE XIII

EFFECT OF DIFFERENT WEIGHTS FOR THE CONSTRUCTED LOSS FOR HSI CLASSIFICATION (%). BEST RESULTS ARE IN BOLD. THE NUMBERS IN PARENTHESES ARE STANDARD VARIANCES (%)

weights	OA	AA	$\kappa \times 100$
1,1,1	$98.68_{(0.57)}$	$99.40_{(0.36)}$	$98.45_{(0.28)}$
2,1,1	$97.82_{(1.13)}$	$98.71_{(1.25)}$	$97.44_{(1.41)}$
1,2,1	$97.60_{(1.68)}$	$98.52_{(1.29)}$	$97.19_{(1.77)}$
1,1,2	$98.84_{(1.13)}$	$99.32_{(0.34)}$	$98.64_{(1.02)}$
4,1,1	$97.77_{(1.91)}$	$98.77_{(0.60)}$	$97.39_{(1.20)}$
1,4,1	$97.39_{(1.24)}$	$98.40_{(1.15)}$	$96.94_{(1.19)}$
1,1,4	$\mathbf{99.05}_{(0.29)}$	$\mathbf{99.42}_{(0.36)}$	$\mathbf{98.88}_{(0.29)}$
1,0,1	$97.95_{(0.41)}$	$98.83_{(0.61)}$	$97.59_{(0.59)}$

weights, we conduct this experiment and obtain the numerical results, as shown in Table XIII. Seven weight combinations are listed in Table XIII. The first weight combination utilizes the equal weight (i.e., all weights are equal to 1). One weight from all three weights is emphasized from the second to the seventh weight combinations. It can be seen that the proposed method performs better when the clustering loss λ_3 becomes large, i.e., the unlabeled data play an important role to improve the classification performance. In this study, we set the weights as [1, 1, 1] for all experiments, though other weights combinations can result in better HSI classification performance.

TABLE XIV

EFFECT OF DIFFERENT CLUSTER NUMBER FOR HSI CLASSIFICATION (%). BEST RESULTS ARE IN BOLD. THE NUMBERS IN PARENTHESES ARE STANDARD VARIANCES (%)

Cluster number	OA	AA	$\kappa \times 100$
5	Failed	Failed	Failed
10	$98.16_{(1.19)}$	$98.99_{(0.95)}$	$97.84_{(1.14)}$
20	$98.36_{(1.13)}$	$98.96_{(0.86)}$	$98.07_{(1.04)}$
50	$98.57_{(1.15)}$	$99.21_{(0.41)}$	$98.33_{(1.12)}$
100	$\mathbf{98.68}_{(0.57)}$	$\mathbf{99.40}_{(0.36)}$	$\mathbf{98.45}_{(0.28)}$
200	$98.68_{(0.86)}$	$99.14_{(0.33)}$	$98.42_{(0.98)}$

TABLE XV

EFFECT OF DIFFERENT CUBE SIZE FOR HSI CLASSIFICATION (%). BEST RESULTS ARE IN BOLD. THE NUMBERS IN PARENTHESES ARE STANDARD VARIANCES (%)

cube size	OA	AA	$\kappa \times 100$
1	$94.06_{(3.65)}$	$96.28_{(2.77)}$	$93.06_{(3.42)}$
3	$98.34_{(0.69)}$	$99.13_{(0.37)}$	$98.05_{(0.92)}$
5	$98.68_{(0.57)}$	$99.40_{(0.36)}$	$98.45_{(0.28)}$
7	$\mathbf{99.33}_{(0.57)}$	$\mathbf{99.65}_{(0.25)}$	$\mathbf{99.21}_{(0.24)}$
9	$98.16_{(0.83)}$	$98.03_{(1.04)}$	$97.85_{(1.23)}$
11	$96.25_{(1.45)}$	$96.56_{(1.58)}$	$96.18_{(1.37)}$

3) Effect of Clustering Number for HSI Classification: In the proposed method, we utilize the K-means clustering method to obtain a cluster label for each unlabeled data. To verify the effect of different clustering numbers, we conduct this experiment and summarize the results in Table XIV, in which we alter the cluster number from 5 to 200. It can be seen that with the increase of the cluster number, the classification performance first increases and then decreases. The reasons are summarized as follows. When the cluster number is small, samples from different classes are prone to be grouped into the same cluster. When the cluster number is

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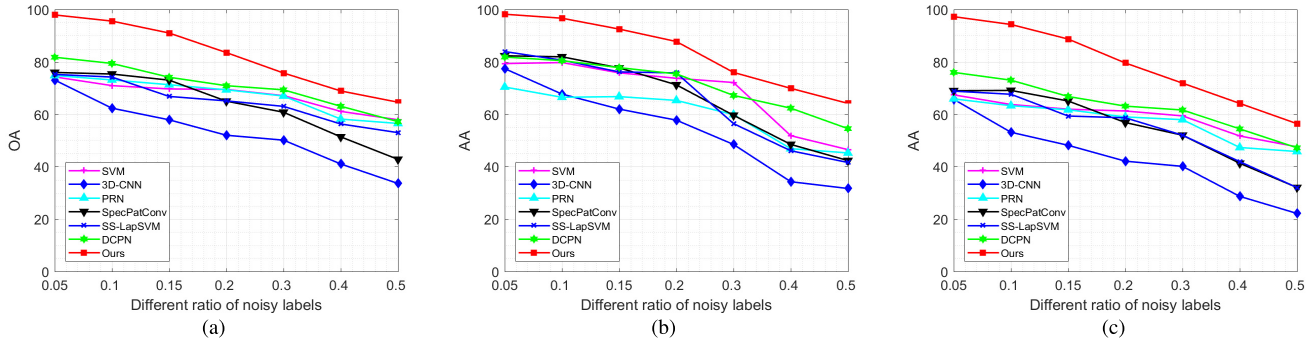


Fig. 10. Classification results of all methods on PaviaU data set varied with different ratio of noisy labels. (a) OA. (b) AA. (c) $\kappa \times 100$.

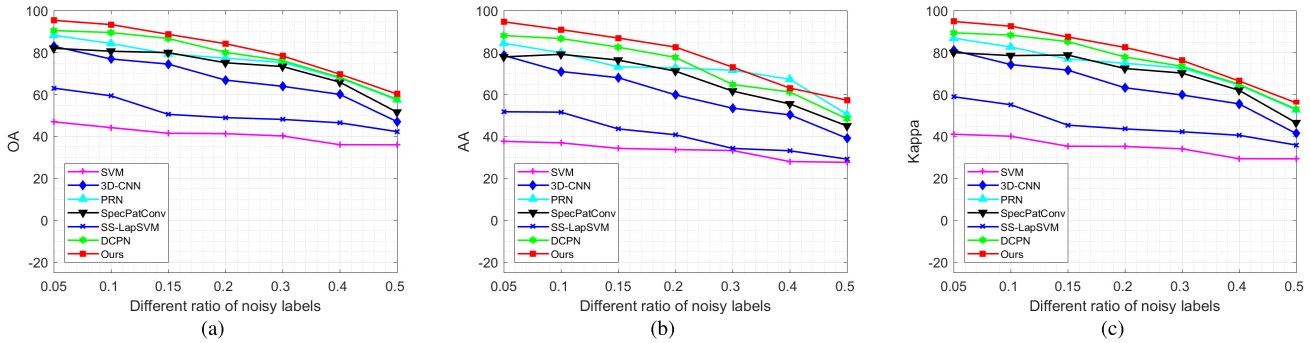


Fig. 11. Classification results of all methods on KSC data set varied with different ratio of noisy labels. (a) OA. (b) AA. (c) $\kappa \times 100$.

TABLE XVI
INFERENCE TIME OF ALL METHODS ON INDIAN PINES DATA SET, IN WHICH 20 SAMPLES PER CLASS ARE UTILIZED FOR TEST

Methods	SVM	3D-CNN	PRN	SpecPatConvnet	SS-LapSVM	DCPN	Ours
Times(s)	1.0488	1.8415	2.9527	0.3721	1.0524	1.9635	1.8508

too large, samples from the same classes will be divided into different small clusters, which will also influence the classification results. Thus, we set the cluster number like 100 in all experiments in this study.

4) Effect of Spatial Size for HSI Classification: In the proposed method, the spatial size of the local cube is set as 5×5. To verify the classification results varied with different spatial size, we conduct this experiment and summarize the results in Table XV. It is noticeable that when the cube size is set as 1×1, the proposed method degenerates into a pixel-level-based HSI classification method. It can be seen that with the increasing of the spatial size, the classification performance first increases and then decreases. With a large spatial size, more neighboring pixels with the same class will be introduced for training, which is helpful to improve the classification accuracy. However, when the spatial size becomes larger, pixels from different categories are prone to be included, which decreases the classification accuracy. In this study, we set the spatial size as 5×5 for all experiments, though another spatial size can result in a better HSI classification performance.

5) Running Time Comparison: The runtime of all methods is listed in Table XVI. The experiments are run on a workstation with Intel Xeon E5-2650 CPU and GeForce GTX 1080ti GPU. It is worth noting that SVM only utilizes the CPU, while other deep learning methods utilize GPU for inference. It can be seen that the runtime of the proposed method is moderate when competing with other methods.

IV. CONCLUSION

In this study, we propose a new unified supervised learning network to jointly take the advantages of both the labeled and the unlabeled data to label data within the HSI. To guarantee the proposed network can be trained with the unlabeled data in a supervised way, we apply a clustering method on the given HSI data beforehand and take the resultant cluster labels as the supervision for those unlabeled pixels to formulate a supervised clustering task. A multitask supervised network is constructed to simultaneously accomplish the formulated supervised clustering task and the conventional classification task. In the proposed network, both tasks share the same basic feature extraction module, and a feature fusion module is also developed to further fuse the high-level feature from two tasks for better feature representation. Experimental results demonstrate that the impressive performance of the proposed method, especially when the label is limited and noisy.

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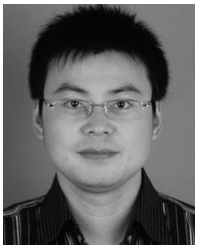
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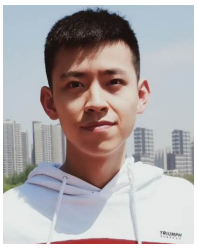


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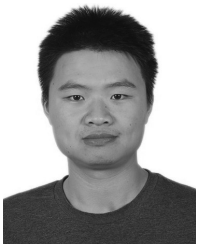
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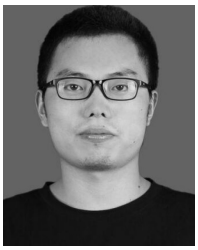
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